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Advanced Optimization of Carbon Usability and Storage Project: An Integrated Approach in Colombia's Magdalena Valley

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Abstract

This study presents an integrated workflow for evaluating and optimizing CO_2 Enhanced Oil Recovery (EOR) and storage strategies in a Colombian field located in the Magdalena Valley. The objective is to assess the feasibility and performance of different injection methods—including Water-Alternating-Gas (WAG), continuous CO_2 injection, and carbonated water injection—under geological and operational uncertainties. The methodology combines compositional reservoir simulation, equilibrium region modeling, and laboratory-validated fluid characterization, supported by a machine learning-assisted history matching process. Key uncertainties such as capillary pressure, relative permeability, and contact depths were rigorously analyzed through sensitivity studies and optimization loops, generating a robust ensemble of predictive scenarios.

Results demonstrate that WAG and carbonated water injection, when combined with optimized injector completions, offer the highest incremental recovery, with improvements of up to 6.5% over the base case. However, carbonated water injection achieves similar recovery with 60% less CO_2 requirement, providing a viable alternative in limited CO_2 supply scenarios. The study also highlights the importance of swelling behavior, vertical injectivity analysis, and pattern-level optimization in achieving dual objectives of oil recovery and CO_2 sequestration. This work advances the understanding of CCUS applications in Latin America and provides a replicable methodology for future integrated EOR and storage projects.

Introduction

Carbon Capture, Utilization, and Storage (CCUS) is a pivotal technology in the global effort to reduce greenhouse gas emissions and combat climate change. By capturing CO2 emissions from industrial sources and either utilizing it in various applications or storing it underground, CCUS can significantly contribute to reducing atmospheric CO2 levels. Enhanced Oil Recovery (EOR) techniques, such as Water-Alternating-Gas (WAG), continuous CO2 injection, and carbonated water injection, play a crucial role in this process by improving hydrocarbon recovery while simultaneously storing CO2.

This study presents a comprehensive evaluation of CCUS methodologies applied in a Colombian field, focusing on the opportunities for EOR and CO2 storage. Various methods, including WAG, continuous CO2 injection, and carbonated water injection, were compared to determining their effectiveness in enhancing oil recovery and storing CO2. The evaluation was conducted in a complex reservoir with approximately 20 equilibrium regions and different fluid compositions.

One of the key challenges in CCUS projects is managing the uncertainties associated with geological formations and fluid behaviors. Recent advances in machine learning have enabled more efficient and accurate optimization of gas injection processes under uncertainty. By integrating machine learning techniques with traditional reservoir simulation models, this study aims to optimize oil recovery and CO2 storage, providing valuable insights for future CCUS projects.

In the context of global CCUS efforts, various studies have highlighted the importance and potential of these technologies. For instance, DaneshFar et al. (2021) conducted an economic evaluation of CO2 capture, transportation, and storage potentials in Oklahoma, utilizing an economic-engineering software tool to integrate infrastructure related to CO2 sources, pipelines, and geological formations. Their study demonstrated the feasibility of CCUS projects by optimizing the process of capturing, transporting, and storing CO2 in geological formations for EOR purposes or deep saline aquifers for sequestration.

Similarly, Lourenço et al. (2023) presented a systematic review of criteria for prospecting potential areas for CCS in Brazil. Their work emphasized the need for comprehensive methodologies to assess the prospectivity of CO2 storage areas, considering the geological characteristics and potential sectors for decarbonization. The study provided valuable insights into the challenges and opportunities for CCS in Brazil, highlighting the importance of integrating scientific research with practical applications to advance CCUS technologies.

In another study, Bao et al. (2023) compared oil recoveries by Huff and Puff gas injection using different gases, including methane, carbon dioxide, hydrogen, and rich gas, in shale oil reservoirs. Their findings demonstrated the efficiency of CCUS and Huff n Puff gas injection in enhancing oil recovery while providing safe storage for CO2 and hydrogen. This dual-purpose strategy aligns with the goals of reducing CO2 emissions and improving hydrocarbon recovery.

Furthermore, Herrera Tellez et al. (2023) conducted reservoir simulations of primary and enhanced oil recovery by Huff and Puff gas injection, and CO2 storage in La Luna Shale, Colombia. Their study revealed that primary recovery from La Luna Shale could be significantly improved by Huff and Puff CO2 injection, while also providing safe and permanent storage for CO2. The concept of geologic containment, where hydrocarbons remain in the same position where they were generated, was crucial for the successful implementation of CCUS in La Luna Shale.

Building on these insights, this study evaluates CCUS methodologies applied to a Colombian field in the Magdalena Valley. The reservoir is particularly complex, with multiple equilibrium regions and variable fluid compositions, which makes it a challenging but relevant candidate for EOR and CO_2 storage assessment.

The methodology integrates compositional reservoir simulation, equilibrium region modeling, and machine learning-assisted history matching to address geological and operational uncertainties. By comparing WAG, continuous CO_2 injection, and CWI scenarios, this study assesses their relative effectiveness in enhancing recovery and maximizing CO_2 storage.

The objectives of this work are to (i) develop a representative static-dynamic reservoir model, (ii) evaluate multiple EOR scenarios, (iii) optimize CO_2 injection by targeting zones with the best injectivity, and (iv) provide recommendations for future CCUS development opportunities in Colombia.

Unlike previous studies in Latin America, which have mainly focused on regional assessments or single-method evaluations, this work introduces a novel integrated workflow that explicitly compares multiple CO_2 injection strategies under uncertainty and quantifies the trade-offs between recovery and storage. This dual-objective approach advances the understanding of CCUS applications in Colombia's Magdalena Valley, making this study the first of its kind in the region and a replicable methodology for similar basins.

Methodology

The methodology for this study involves several key steps to ensure a comprehensive evaluation of CCUS methodologies and EOR techniques in the Colombian field. The process is divided into the following stages:

Initialization of the Model

Data Collection. The first step involves gathering existing geological and petrophysical data, including relative permeability curves and capillary pressures. This data is crucial for constructing accurate models of the reservoir. New capillary pressure curves were generated for each rock type based on the grid of capillary pressure. These curves were designed to show higher irreducible water saturation as rock quality improves, thus more accurately reflecting the characteristics of each rock type.

Equilibrium Regions. Using the collected data, equilibrium regions were defined with Free Water Levels (FWL) based on saturation curves and transition zones, the model comprises 2 main fluid zones and around 20 equilibrium regions. This step ensures that the model accurately represents the different fluid distributions and transitions within the reservoir. The figure shows the correction of the initial water saturation of the reservoir. The red points represent the initial water saturation data, while the green bars indicate the refined values after adjusting the capillary pressure curves. The black curve represents the original saturation from the log for the first well. This refinement process was essential for improving the initialization of the model, taking into account the different equilibrium regions and the compilation of rock property data

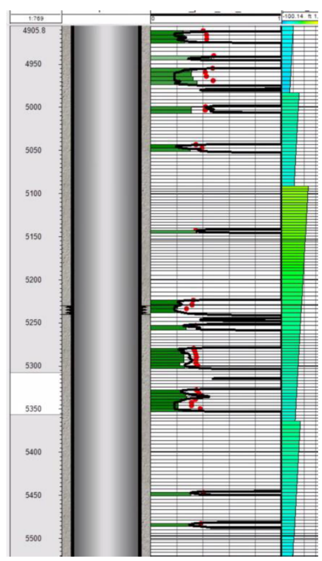

Figure 1—Water saturation log for initialization model.

Dynamic Model Evaluation

EOS Characterization. Equation of State (EOS) were characterized for the field as 7 components Peng Robison equation used to model the fluid behavior. Initial fluid conditions were tested through flash tests to ensure the accuracy of the model.

Swelling Tests: Swelling tests were conducted to verify the representativeness of CO2 in the reservoir at different fluid compositions. These tests were performed for the two main formations to understand how CO2 interacts with the existing fluids and to assess its impact on viscosity and fluid behavior and the results were compared with laboratory analysis. The image illustrates the comparison between the laboratory-validated results and the Pedersen viscosity models, alongside the simulation outcomes for four distinct fluid compositions. These comparisons were conducted under varying pressure conditions and different gas-oil ratio (GOR) levels. The continuous blue lines represent the Pedersen fluid model validated by laboratory tests, while the red dashed lines depict the simulation results.

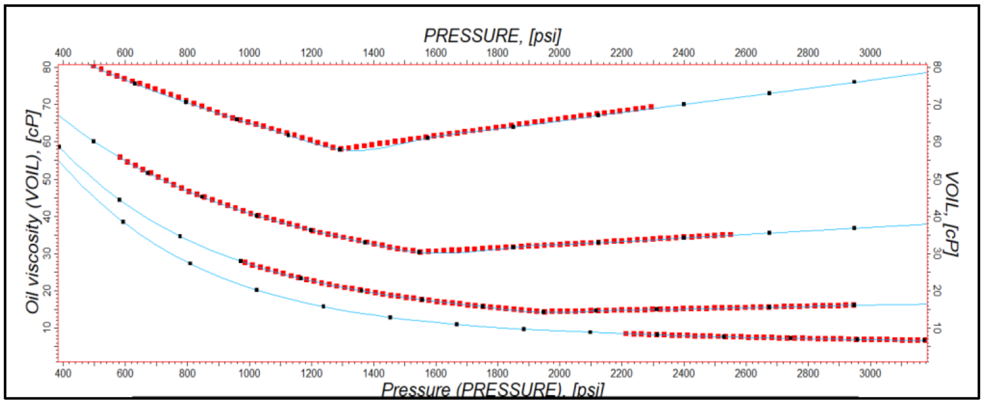

Figure 2—Comparison of Viscosity results: fluid model validation vs. Simulation for 4 Fluid Compositions.

A verification of the events was conducted with the production history to validate the concordance between both. Additionally, pseudo wells were created to load static pressure data, allowing us to use this pressure data as one of the drivers in the objective function to reach the goal for the history matching.

Through the analysis of pressures and water production, equilibrium regions with a likelihood of connection to aquifers were identified, which were modeled analytically.

Integrated Static-Dynamic Workflow.

- Uncertainty Management: A comprehensive workflow was created to integrate static and dynamic models for managing uncertainty. This workflow replicates changes from property modeling to initialization of the simulation model and includes several uncertainty variables such as:
    - Depth of contacts
    - Relative permeability parameters
    - Numerical aquifer-related variables
    - Maximum capillary pressure
    - Pore volume multiplier at grid edges (connection with neighboring areas)
    - Clay probability to represent communication levels between sands
- Assisted history match with Machine Learning: This section details a machine learning-assisted history matching process, employing an iterative and optimized approach to identify the best combinations of uncertainty parameters in a simulation model. Initially, 40 relevant uncertainty variables were selected from all those considered, considering time constraints and the requirement for simulation runs to train the model. Through a sensitivity analysis using the Plackett-Burman and Monte Carlo design of experiments. The results were used to train a machine learning model based on Random Forest, which enabled the generation of 10,000 simulations using random sampling and analysis based on SHAP (Shapley Additive Explanations) values. This analysis identified the top 100 runs based on their impact on the objective function. The best runs were analyzed to define ranges for the key variables and retrain the model, iterating the process. Finally, the 50 best-optimized results were obtained, focusing on maximizing the objective function

Forecast Scenarios.

- Scenario Evaluation: Considering the field's history, the characteristics of its formations, and the corporate and state objectives, multiple EOR scenarios were evaluated during the forecast phase, including:
  - Base Case (no further action)
  - Water injection
  - Continuous CO2 injection
  - Water-Alternating-Gas (WAG) injection
  - Carbonated water injection

To compare all these cases, the following criteria were defined for all prediction strategies:

- Shut-in completion with the highest GOR upon reaching a well GOR (Gas-Oil Ratio) limit.
- Shut-in completion with the highest water cut upon reaching a water cut of 98%.
- Total injector wells 2 (current injectors wells)
- Prediction period: 20 years
- Producer wells controlled by BHP at the end of the history match run
- WAG case using Killough 1976 hysteresis model (2 phases)
- All cases using Stone 2015 Isenthalpic flash model
- Injector wells with a maximum BHP based on geomechanical risks

Uncertainty Runs and Optimization

- Uncertainty Evaluation: To evaluate the EOR scenarios, compare them with the base case, and select the best case, 100 uncertainty cases were run for each prediction scenario. An objective function was included to maximize oil production while minimizing CO2 recycling, if applicable. Extreme values were defined for Monte Carlo runs.
- Injector Well Optimization: The option to optimize injector well completions was evaluated to observe their impact on recovery. An automated workflow was built to test different combinations and evaluate which results in higher recovery and lower CO2 recycling. Cases were run with each equilibrium region open separately, keeping the rest closed. The entire region above the oil-water contact was perforated.
- Injector parameters optimization: The uncertainty runs considered variations in several injection parameters to identify optimal cases based on the best combination of variables. For continuous injection scenarios (water or CO2), variations in injection rates were analyzed. For Water-Alternating-Gas (WAG) injection scenarios, both the injection rates and the injection cycles (rates and periods) were evaluated. Additionally, for carbonated water injection scenarios, different percentages of CO2 in the carbonated water were tested. These variations allowed for the selection of optimal cases by determining the most effective combination of injection parameters

Results

After developing a calibrated and representative model, the focus shifted to evaluating the Enhanced Oil Recovery (EOR) and storage processes. The results were analyzed considering uncertainties, variables, and test outcomes

Injection Optimization

To optimize the injection process, uncertainty runs were conducted for each scenario, including Water-Alternating-Gas (WAG), continuous CO2 injection, and carbonated water injection (CWI). These runs involved varying injection rates, cycles, and the proportion of CO2 in the carbonated water. The goal was to identify the optimal combination of parameters that maximize oil production while minimizing CO2 injection.

For each scenario, a series of uncertainty runs were performed to explore the impact of different injection parameters. In the WAG scenario, both the injection rates and cycles (rates and periods) were varied. For the continuous CO2 injection scenario, variations in injection rates were analyzed. In the CWI scenario, different percentages of CO2 in the carbonated water were tested to determine the most effective combination.

In the scatter plots, each point represents a different case, showing the relationship between injection and production. The optimal cases are highlighted with stars in the graphs.

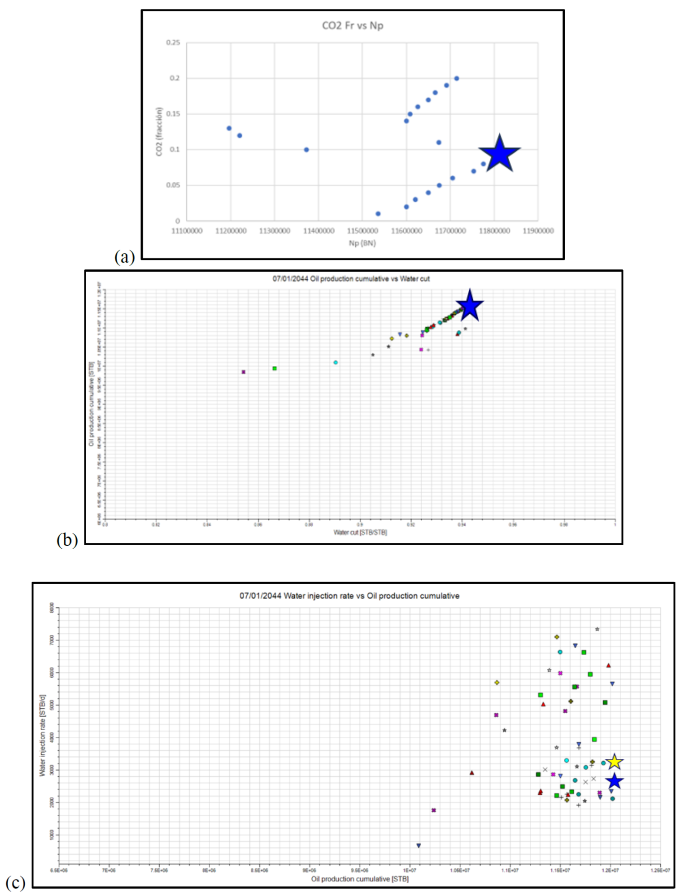

Figure 3—scatter plots for uncertainty and optimization analysis for 3 different forecast scenarios.

Optimization of Injector Completions

One of the key factors affecting gas injection is the recirculation of gas through high-permeability channels. This necessitates the optimization of existing wells to select zones where injection will result in the most significant production increase. To address this, uncertainty runs were conducted for each scenario identifying the optimal combination of parameters that maximize oil production while minimizing CO2 injection.

A scatter plot was used to visualize the field-level increase and its relationship with the cumulative CO2 injected. The plot identified cases where, despite injectivity, there was no significant field-level increase. Conversely, it highlighted zones where minimal CO2 injection resulted in a substantial increase in cumulative oil production (Np). Each curve represents a different zone in the reservoir.

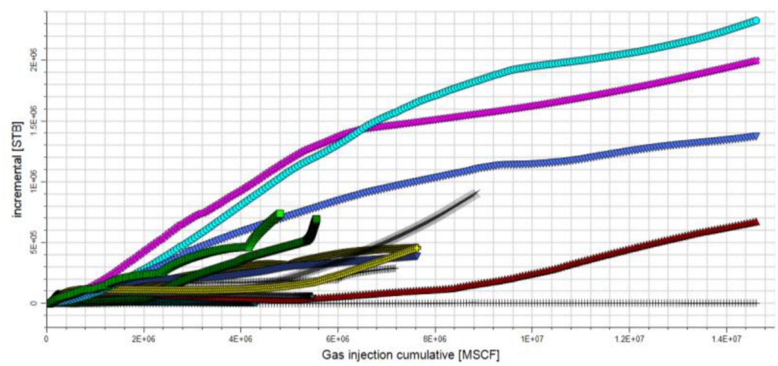

Figure 4—field incremental vs gas injection cumulative for different reservoirs zones.

Based on these results, uncertainty and optimization runs were conducted, assigning an 80% weight to cumulative Np and a 20% weight to cumulative CO2 produced

The image shows two yellow stars and one blue star. The blue star represents the best case for EOR projects considering the weighted mentioned before, while the lower yellow star represents the best case for Storage projects. The line formed between the two yellow stars indicates the trend towards the optimal injection case, where moving upwards increases the impact on EOR projects while also achieving CO2 storage in the reservoir.

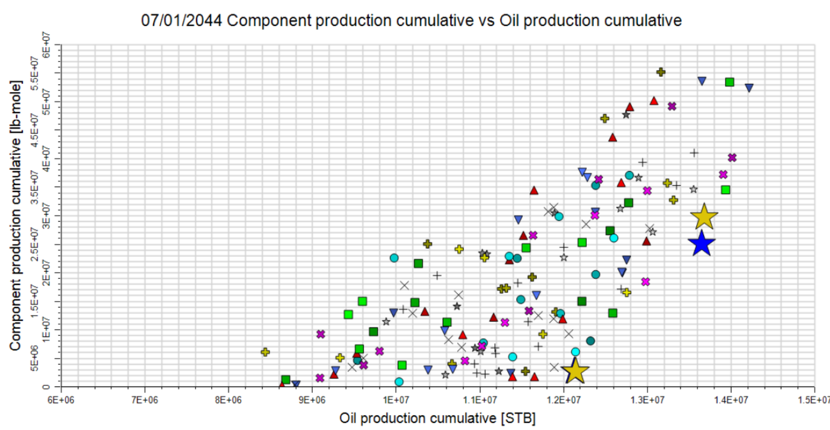

Figure 5—CO2 production cumulative vs oil production cumulative for every uncertainty scenario and marked scenarios.

Optimal zones were selected from each injector for the scenario comparison between different strategies. Once the results of each prediction scenario were obtained, a total production cumulative graph was created to see the impact of each scenario on oil production for the entire area and specifically for patter area.

The scenario with the highest cumulative production was the WAG case, followed by CWI, both with optimized perforations. The incremental recovery factor was calculated for the two mentioned areas.

It was observed that the WAG case with new perforations in the injectors would generate an additional recovery of 6.5%, making it the best EOR scenario for the reservoir conditions in this area. However, despite being the best option, the WAG scenario requires 60% more gas injection compared to the CWI scenario, which achieves similar recovery levels. This makes CWI a viable alternative in situations where CO2 availability is limited

Comparison of WAG and CWI Performance

To further evaluate the effectiveness of the studied injection strategies, a direct comparison between Water-Alternating-Gas (WAG) and Carbonated Water Injection (CWI) was performed (Figs. X and Y). Both cases achieved a very similar ultimate oil recovery, approximately 17 MMSTB, confirming that both methods can unlock significant additional reserves under the studied conditions.

However, the difference in CO_2 utilization between the two strategies is remarkable. The WAG case required about 2.4E+07 MSCF (≈ 24,000 MMscf) of cumulative CO_2 injection, while the CWI case achieved comparable recovery with only 5697 MMscf. This indicates that CWI can reduce CO_2 consumption by more than 60%, highlighting its potential advantage in scenarios where CO_2 supply is limited.

The observed inefficiency in the WAG cycles is mainly attributed to preferential gas flow through high-permeability channels, which leads to early gas breakthrough and higher recycling requirements. In contrast, CWI ensures a more uniform sweep, as CO_2 is dissolved in the injected water, mitigating preferential flow paths.

Operationally, the optimized WAG design in this study consisted of 100 days of water injection followed by 140 days of gas injection, repeated cyclically. While this strategy was effective in sustaining production rates (up to 1156 STB/d peak), the higher CO_2 demand raises concerns about project feasibility in regions such as Colombia, where CO_2 availability and infrastructure may become limiting factors.

From a storage perspective, WAG favors a larger amount of injected CO_2 retained in the subsurface; however, when considering both recovery efficiency and resource availability, CWI emerges as an attractive and practical alternative. Its ability to balance oil recovery with reduced CO_2 requirements makes it particularly relevant for CCUS deployment in Colombia, where minimizing CO_2 sourcing and transportation costs is critical for project success.

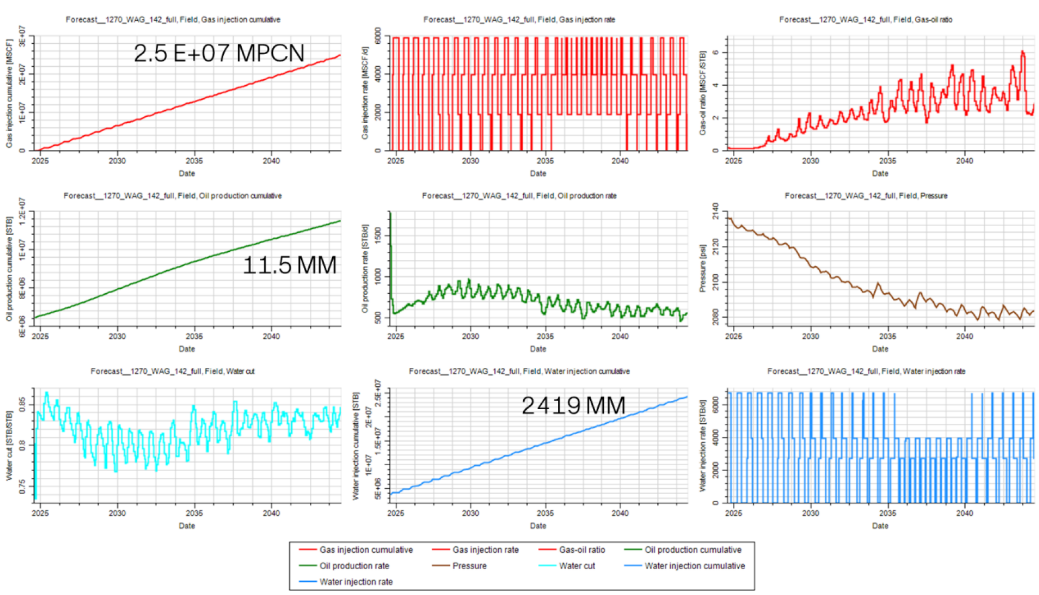

Figure 6—Performance of WAG scenario showing cumulative oil production, CO_2 and water injection, pressure behavior, and injection cycles (100 days water, 140 days gas).

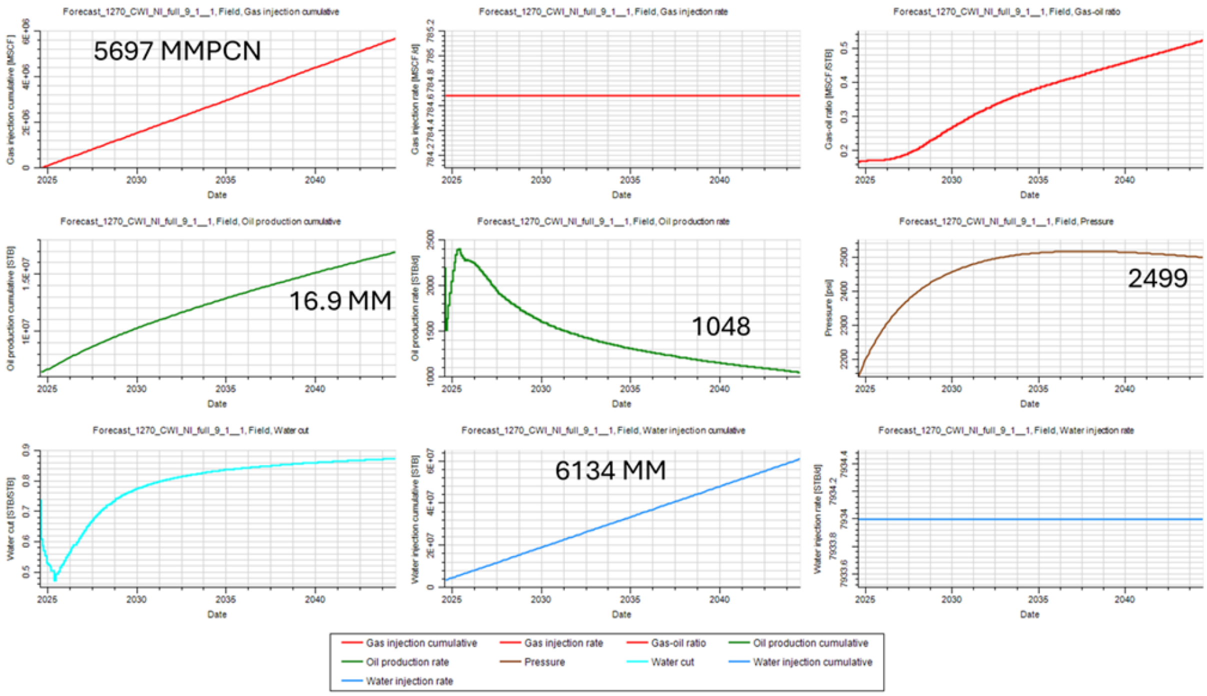

Figure 7—Performance of CWI scenario showing cumulative oil production, CO_2 and water injection, pressure behavior, and stabilized continuous injection rates.

Swelling Evaluation

To evaluate swelling in the reservoir, sections of the reservoir was evaluated, an example is showing in the image below with a cell near to a producer well. Pressure, total CO2 molar fraction, oil viscosity, immobile CO2 mass, oil saturation, and CO2 mass in water were plotted.

For the three scenarios, as the total CO2 molar fraction increases, oil viscosity decreases, reflecting CO2 swelling in the oil. Additionally, an increase in immobile CO2 and a slight increase in water CO2 were observed. Pressure initially increases and then shows a slight decline due to high well production without being dominant. To observe the impact of EOR scenarios on residual oil saturation, the static Sor from the history match case with a value of 18% was compared with the oil saturation at the end of the prediction cases with around 2% reduction in different CO2 scenarios.

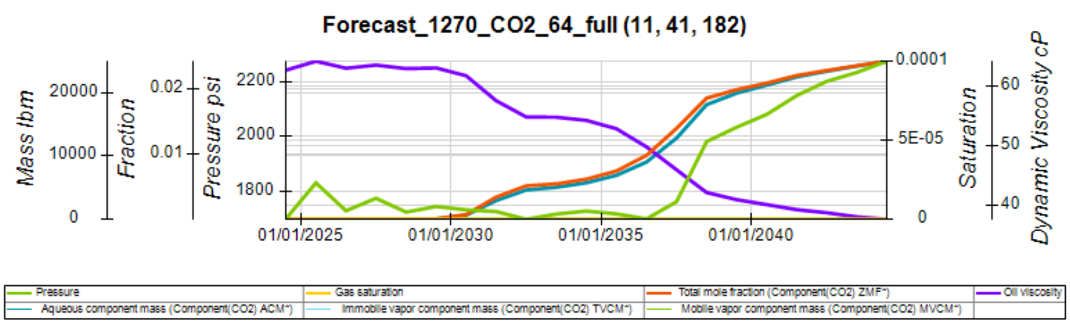

Figure 8—Temporal Evolution of CO2 Molar Fraction, Oil Viscosity, and Pressure.

Areal and vertical injectivity

To identify the vertical injection fractions in the well, simulation PLTs and saturation maps for each region were reviewed. For areal distribution, flow lines were run to determine allocation factors between nearby wells and the extent of CO2 injection into adjacent areas. This approach allows for understanding the impact on neighboring areas across different zones. In the image below, you can see PLTs for the wells and saturation maps for three zones

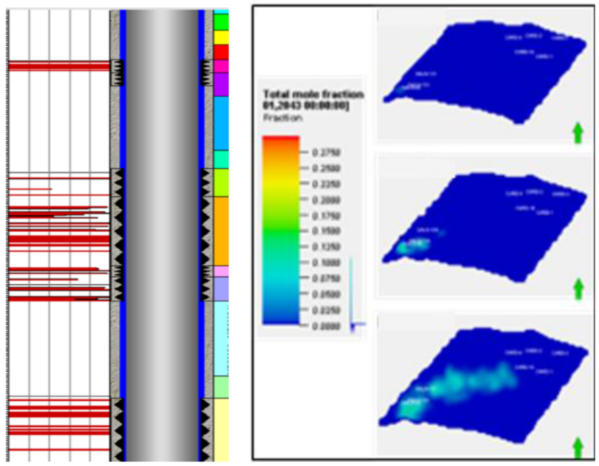

Figure 9—PLT para pozo inyector y mapas de saturación de CO2.

Conclusions

A black oil simulation model was successfully converted into a compositional model through a thorough review of critical parameters representing the fluid behavior in the reservoir for dynamic evaluation

Capillary pressure and relative permeability curves were adjusted to better reflect rock quality and improve movable water control.

Sand probability flows, property calculators, and aquifers were analyzed and optimized to create an ensemble of uncertainty simulations capable of representing reservoir behavior. These simulations were used to evaluate and compare multiple EOR scenarios with the base case, allowing for the selection of the best scenario using an objective function. This approach ensures accurate future predictions and effective evaluations of EOR and CO2 storage scenarios.

The optimization of injector well completions showed that Water-Alternating-Gas (WAG) injection and carbonated water injection (CWI), both with optimized perforations, would generate the highest incremental recovery. Depending on the availability of CO2 and water, it is possible to select one method over the other. Both methods present potential opportunities for recovery increases between 3% and 6% compared to the no further action scenario, without increasing the number of injectors or affecting the current conditions of the producer wells

In summary, the project successfully converted and optimized a black oil simulation model into a compositional model. Through an iterative and detailed process, critical parameters were adjusted, multiple prediction scenarios were defined and evaluated, and the best EOR case was identified to maximize oil production and minimize CO2 recirculation.

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